Which Way Does Fairness Go? The Selection Rate Predicts Whether an Attribute-Blind Constraint Gives or Takes

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Abstract*—A group-fairness constraint requires two groups' selection rates to be close, but it does not say how the gap should close: the disadvantaged group can be lifted, or the advantaged group can be pulled down. Both outcomes satisfy the constraint, and the metric that certifies it reports the same number in each case, which means a deployment team cannot tell from its own dashboard which of the two it is about to cause. We show that the direction is predictable in advance from a quantity every deploying organisation already has: the rate at which its current model says yes, read against that population's crossover. Across 57 independent populations covering six decision domains and seven data sources, an attribute-blind parity constraint withdraws favourable decisions when the baseline selection rate sits below a crossover, and extends them when it sits above; on UCI Adult five in-processing methods each shrink the pool by 7.9–22.1%, while on mortgage lending — one two-state market, measured three overlapping ways — the identical constraint grows it. The crossover's location is population-specific — located values span 0.28 to 0.65, and the fixed 0.54 prior is a fallible substitute for measuring one's own: sealed at 9 of 10 on one cohort of never-measured populations (best constant 6; binomial tail 0.046 against a guesser, but only $p \approx 0.19$ paired against the constant itself) — a cohort whose arms all reach their rates through the income cutoff, so the pass cannot yet separate the selection rate from label rarity — and 5 of 10 post-hoc on a second cohort of larger states, every miss agreeing with that population's own crossover. The sealed cohort's rule, populations and pass mark were committed to version control before any run; an earlier sealed attempt that carried an in-sample refinement failed at four of eight, and we report both outcomes because the difference between them is what a sealed test exists to expose. A caution that generalises beyond this paper: four 2022 populations initially appeared to *invert* the relationship, and diagnosis traced this substantially to the fixed nominal $50,000 label sliding down an inflated income distribution — a fixed nominal outcome definition is a different task each year, so labels should be real-anchored and any deployment should measure its own current data rather than transport results across vintages. The relationship survives a change in how the rate is varied, a twenty-five-fold range of constraint tolerance, a boosted-tree learner and a protected attribute swapped from sex to race, but it disappears under post-processing ($r = -0.024$ against $+0.585$), which reads the protected attribute at decision time. That boundary is where contemporaneous theory says the direction stops being distributional and becomes determined, and the theory's prediction for that regime holds in 27 of 27 populations, nine pre-registered. The practical outcome is a procedure: when the deployed model's viable operating band is wide enough — which the procedure itself checks first — a team can locate its own crossover by sweeping that model's threshold, using no new data.

***Index Terms*—algorithmic fairness, levelling down, demographic parity, in-processing, selection rate, disparate impact**

I. Introduction

A fairness constraint says that two groups' rates should be close, but it does not say how that closeness should be reached. The gap can close because the disadvantaged group is lifted, or because the advantaged group is pulled down, and while both directions satisfy the constraint, they are morally very different outcomes on most — though, we note, not all — accounts of what the constraint is for; at minimum, anyone imposing one deserves to know which they are getting.

This ambiguity is already well established. Mittelstadt *et al.* [2] made levelling down the centre of a well-known critique; Maheshwari *et al.* [4] observe that it "often goes unnoticed in the overall performance of the model"; and Ferry *et al.* [3] built an auditing framework precisely because an aggregate score reports neither how many people were affected nor in which direction. We take all of that as given.

What the literature leaves open is the question a deploying team actually faces: **when** does each direction happen? A team cannot act on "this sometimes occurs", but it can act on a rule that says which way its own system will move before it moves, which is what this paper provides and tests.

A. Scope, stated first

This is a paper about **attribute-blind in-processing** constraints, meaning the regime in which the protected attribute is unavailable at decision time. This is the regime most deployed systems occupy: in the United States, per-group decision thresholds face direct disparate-treatment exposure (*Ricci v. DeStefano* [18]; in credit, ECOA and Regulation B prohibit attribute-based terms outright), and commentators read the doctrine the same way [10], [17]. Two honest qualifications. First, the barrier is jurisdiction-specific, and the two major jurisdictions draw it in different places: in the US even *training-time* use of the attribute — which every in-processing method here involves — is legally unsettled, while in the EU the split runs the other way: *decision-time* use of special-category data faces GDPR Article 9 and direct-discrimination doctrine, but training-time use for bias detection and correction in high-risk systems is expressly permitted, under safeguards, by Article 10(5) of the AI Act — and for race and ethnicity the blind regime is often simply factual, because most member states collect no such attribute at all. Second, because of that split we motivate the regime as the one systems occupy, not as one any law cleanly compels. Under post-processing, which does read the attribute, the effect we report disappears (Section VII). We treat this not as a weakness of the finding but as the boundary the theory draws, and we verify it from both sides.

B. Contributions

1) **An empirical characterisation of a conditionality that theory predicts.** Contemporaneous theory [5] establishes that in the attribute-blind regime the direction is distribution-dependent. It contains no experiments and its conditions are ordering relations on score regions rather than measurable quantities. We supply the measurement across 57 populations and six domains, and find that under our probe estimator its stated conditions — which are *sufficient, not necessary* — could not be certified on any of 26 real arms (Section VII), while a relaxed form of the same ordering agrees with the measured direction on 24 of 26 and correlates with our predictor at $r = +0.935$: the selection rate operationalizes the theory's quantity. The direction rule itself is *sealed*: committed before ten populations this work had never measured, it scored 9 of 10 against a best constant of 6 (Section VIII-E) — with the confound stated where the score is: every arm of that cohort reaches its rate through the income cutoff, so the pass does not yet separate the selection rate from label rarity, and the deconfounding cohort is specified in the same section. On a second, post-hoc cohort of larger states the fixed prior scored 5 of 10, each miss agreeing with that population's own measured crossover.
2) **Two predictive claims, distinguished by what they require.** The paper contains a *sweep-conditional* claim — run the audit on your own data and, where the response is monotone, the rate against your own measured crossover predicts direction — and a weaker *transported-prior* claim, the only one testable before any measurement, whose record is the sealed 9 of 10 followed by 5 of 10. Varying the selection rate by moving a label threshold cannot alone distinguish "the rate predicts" from "task difficulty does"; holding the task fixed and moving only the decision rule separates them across the mid-range, and we say *predicts*, never *causes*: no pooled slope transfers across populations (Section VIII).
3) **A procedure, not an observation.** The same design lets a practitioner locate their own crossover on a deployed model. Three overlapping estimates of the mortgage crossover agree, all from one two-state market.
4) **Boundaries stated explicitly rather than left for review to find**, including a computable test for when the procedure cannot be used at all.

II. Related Work

Whether levelling down is universal. A 2026 theory paper [5] proves in a population-level Bayes framework that the answer depends on the deployment regime. Where the protected attribute is available at decision time, fairness "necessarily (weakly) improves outcomes for the disadvantaged group and (weakly) worsens outcomes for the advantaged group". Where it is not — the attribute-blind regime, which is ours — "the impact of fairness is distribution-dependent: fairness can benefit or harm either group . . . leading to either leveling up or leveling down." **We therefore do not claim the conditionality itself.**

Levelling down, and the remedy. Mittelstadt *et al.* [2] argue that fairness interventions frequently equalise by degrading the better-off group, and *also propose the remedy*: minimum rate constraints requiring "that every group has, at least, a minimal selection rate, precision, or recall". We claim neither the observation nor the remedy as novel. Our contribution relative to that work is narrow and checkable: a scope condition they do not have; in-processing rather than post-processing, so no model reads the attribute when predicting; a population-level floor stacked *with* parity rather than a per-group floor replacing it; and held-out replication where they report training-set results.

Auditing individual impact. Ferry *et al.* [3] propose a framework whose first three dimensions are impact size, change direction and decision rates. That is the decomposition we use throughout, and we claim no part of it. What we add is what those axes report at scale: the change-direction dimension is not a property of the method being audited but of the population it is applied to.

The shape of the optimal fair classifier. Corbett-Davies *et al.* [10] show that for several fairness definitions the optimal constrained classifier takes the form of group-specific risk thresholds. This gives a *bound* rather than another baseline: it predicts that levelling down is not an artifact of the reduction's search, because the optimal constrained classifier withdraws in the same way.

Remedies that avoid harm. Minimax group fairness [11] minimises the worst group's error rather than equalising anything, and is the natural alternative to a parity constraint for a team whose concern is harm rather than equality. We benchmark against it rather than only against the unconstrained reduction.

The selection-rate axis. The nearest prior work is Goethals *et al.* [8], which studies the cost of fairness as a function of available budget and reports that "the level of available resources significantly influences this cost, a factor overlooked in previous evaluations". The difference is structural: in their setting positive decisions are a fixed resource, so the total is constant by construction and the directional effect we measure cannot arise.

Fairness gerrymandering and dataset monoculture. Kearns *et al.* [6] show constraints on marginal groups can be satisfied while subgroups are badly treated. Ding *et al.* [7] argue the field should stop drawing conclusions from UCI Adult and supply ACS-derived replacements. Notably they also *build the knob this paper turns*: the $50,000 threshold "was chosen so that this dataset can serve as a replacement to UCI Adult, but we also offer datasets with other income cutoffs". They do not connect it to the direction of a fairness intervention — their text contains no occurrence of "selection rate", "acceptance rate" or "leveling down" — so the instrument existed and the question was not asked of it. We replicate across their populations and then find that is *not sufficient*, because they share one survey instrument.

III. Setup and Method

Data. Fifty-seven independent populations across seven data sources: UCI Adult (45,222 rows); ACS Income [7] across forty US states, ten further state-years (2014, 2019 and 2022) and two protected attributes; HMDA 2018 mortgage filings for Mississippi and Louisiana (79,104 and 123,843 records), where the outcome is the *lender's* approval — action-taken codes 1 (originated) and 2 (approved, not accepted) against 3 (denied), with withdrawn, incomplete, purchased and preapproval records excluded as not being a completed approve-or-deny — features restricted by allow-list to what exists when the application is filed, the race arm White vs. Black applicants, the sex arm Male vs. Female (joint applications excluded), and loan purposes as sub-populations; COMPAS [9] (5,278 rows on the race arm after ProPublica's documented filter); LSAC bar passage (20,798 rows); the Dutch national census of 2001 (60,420 rows); and Taiwanese credit-card default records of 2005 (30,000 rows). A *population* here is one such unit — a state, cohort or census — and *independent* means disjoint person samples: no person appears in two of them, and arms within a population share people, so however many arms a population contributes, it is counted once (Table I); the seven data sources remain the honest count of independent *instruments*. Three survey-methodology conventions are deliberate and inherited from the folktables benchmark [7]: ACS person records are used *unweighted* (no `PERWT`), incomes are nominal `PINCP` without the `ADJINC` adjustment, and household clustering is not modelled, so every "selection rate" here is a property of the unweighted sample rather than a design-consistent state estimate, and seed-resampled intervals are anti-conservative. Two further PUMS facts bound what the labels can mean: state topcodes sit far above every cutoff used here, so topcoding is benign; a fifth or more of PUMS incomes are hot-deck imputations, a share that rose in pandemic vintages, and no analysis here excludes allocation-flagged rows — an untested candidate contributor to any income-label result, noted rather than resolved. A weighted re-analysis is reported as ongoing work.

COMPAS, LSAC and the Dutch census were added to leave the sources the finding was formed in. The Dutch census carries a between-group gap of 0.298, roughly twice Adult's, making it the test of the standard alternative explanation. LSAC was chosen because bar passage is naturally generous (base rate 0.890).

Definitions, in one place. An *arm* is one population under one configuration (a decision threshold, a label cutoff, a tolerance, a learner). The *selection rate* r is the fraction of the test split the baseline model marks favourable. The *pool change* Δpool is the percentage change, baseline to mitigated, in the number of favourable decisions; *up/extension* and *down/withdrawal* name its sign. A population's *crossover* is the rate at which its pool change changes sign, reported as the bracket between the highest rate that still shrinks the pool and the lowest that grows it. The *viable band* is the span of selection rates over which the baseline stays above the trivial predictor. The *exchange rate* is favourable decisions destroyed per favourable decision created, an unweighted descriptive count.

Protocol. The baseline learner is an L2-regularised logistic regression on one-hot-encoded categorical and standardised numeric features; one section swaps in a boosted tree and says so. Each arm draws a 70/30 train/test split, stratified on group $\times$ label, re-drawn per seed. Five seeds per arm unless stated (the replication checks use twenty); every per-arm quantity in this paper is a mean over its arm's seeds, and no single-seed number is comparable to the tables.

Convention. $y = 1$ is always the favourable outcome. This required inverting COMPAS's recorded target, which encodes recidivism, and declaring *Female* the advantaged group on its sex arm, since women in that cohort reoffend less. Both are enforced by automated checks, because either error produces every directional result sign-inverted without raising.

Mitigation. `ExponentiatedGradient` [1] at $\varepsilon = 0.01$ unless stated (ε is the library default and bounds the *training-time* empirical moment violation of the randomized classifier, so held-out gaps such as Table II's 0.018 can lawfully exceed it; the tolerance sweep spans 0.002–0.05, a 25$\times$ range). Fairlearn 0.14.0 throughout, default `ExponentiatedGradient` parameters beyond ε. No model reads the protected attribute at prediction time except where post-processing is explicitly under test.

Exclusion rules. An arm is discarded if the baseline demographic-parity gap is below 0.05, or if the baseline classifier's accuracy falls below $\max(p, 1-p)$ — the accuracy of always predicting the more common label. The second rule matters more than it sounds (Section X). Provenance, because exclusion rules can select on outcomes: both rules were developed *during* the exploratory phase, in response to failures the ledger records, so retained-arm correlations from that phase are conditioned on in-sample-tuned filters and the sensitivity table reports the dependence; every sealed test

TABLE I

WHICH POPULATIONS ENTER WHICH ANALYSIS. AN *arm* IS ONE POPULATION UNDER ONE CONFIGURATION; POPULATIONS ARE COUNTED ONCE HOWEVER MANY ARMS THEY CONTRIBUTE.

Analysis	Populations (arms)
Ablation; who pays	Adult (5 methods $\times$ 5 seeds)
Rate–people divergence	10 pops. (19 arms, 2 attributes)
Label route (income cutoffs)	AL sex; AL, MS, OR, UT race (4 each)
Natural transition	HMDA MS+LA (5 loan purposes)
Threshold route, dense sweeps	Dutch, SC, COMPAS, AL, KY (12 pts.)
Tolerance, 25$\times$ range	AL, CT, KY, OR, SC
Boosted trees	the same five states
Intersectional	Adult + 10 populations
Post-processing regime	17 populations ($r = -0.024$)
Attribute-aware direction	27 of 27, of which 9 pre-registered
Relaxed ζ ordering	26 arms from 15 populations
Crossover intervals	4 non-lending + 3 lending estimates
Sealed test 1 (failed)	8 scored + Taiwan, all fresh
Re-seal (holds)	10 fresh ACS states
Residual campaign (underpow.)	10 fresh states (76 arms)
Sealed magnitude (failed)	NY + TX (16 arms)
Cross-year shape seal (failed)	8 fresh state-years (70 runs)
Attr.-independence seal (failed)	6 race arms (48 runs)
Selection-rate floor	10 pops. (19 arms)
Cross-task shape seal (failed)	6 task arms, 4 states' persons; not indep.

TABLE II

FIVE IN-PROCESSING METHODS ON UCI ADULT, FIVE SEEDS. EVERY ONE REACHES PARITY, AND EVERY ONE SHRINKS THE POOL.

Method	Acc.	DP	EO	DI	Δpool
Unconstrained	0.847	0.186	0.095	0.300	—
ExpGrad (DP)	0.828	**0.018**	0.280	0.895	-20.5%
GridSearch (DP)	0.827	**0.015**	0.304	0.986	-22.1%
Adversarial deb.	0.826	0.020	0.250	0.889	-14.8%
Prejudice remover	0.837	0.065	0.188	0.686	-10.2%
ExpGrad (EO)	0.836	0.107	**0.032**	0.521	-7.9%

from the re-seal onward carried both rules frozen in its pre-committed protocol.

Accounting. Table I states which populations enter which analysis, because three counts in this project's own history turned out to be arm counts masquerading as population counts, and a reader should not have to reconstruct the denominators.

IV. PARITY IS REACHED BY WITHDRAWAL — IN SOME POPULATIONS

Table II gives the ablation. Every method reaches its target: demographic parity falls from 0.186 to as little as 0.015 for under two accuracy points, and this holds on a linear model and on an adversarial one alike. The mitigation itself works, which is why we treat it as background rather than as a contribution.

What changes alongside is the part the metric does not report: **every one of these methods shrinks the pool**, by 7.9% to 22.1%, and not one of them closes the gap primarily by extending decisions to the disadvantaged group. Eighteen of nineteen survey arms show the same pattern, which suggests this is a property of the populations rather than of any one method.

TABLE III

WHO PAYS, UCI ADULT. THE RATE-LEVEL VIEW AND THE PEOPLE-LEVEL VIEW OF THE SAME INTERVENTION DISAGREE SYSTEMATICALLY.

Method	Share (rates)	Share (people)	Exchange
ExpGrad (DP)	0.575	**0.742**	2.68
GridSearch (DP)	0.575	**0.743**	2.69
Adversarial deb.	0.508	**0.700**	1.96
Prejudice remover	0.498	**0.673**	2.04
ExpGrad (EO)	0.531	**0.660**	1.92

A. Rates understate what is happening to people

The decomposition of [3] splits the closed gap into the part paid by the advantaged group losing ground and the part gained by the disadvantaged group. Measured in *rates*, that split is close to even — 0.50 to 0.58 of the closure comes from withdrawal (Table III).

Measured in *people* the split is not even. Between 0.66 and 0.74 of the individuals whose decision changed are members of the advantaged group losing a favourable one. The reason the two views disagree is that equal-looking rate changes fall on groups of unequal size, so the rate-level view — which is what a fairness dashboard reports — understates the burden counted in individuals, and it does so in every method tested. The divergence between the two views is itself predictable from a cross-flow diagnostic, at $r = +0.885$ across populations.

The **exchange rate** shows the same thing without needing a share: ExpGrad under demographic parity destroys 2.68 favourable decisions for every one it creates.

The normative baseline, stated as the convention it is. Throughout, "withdrawal", "who pays" and the exchange rate are counterfactual-comparative quantities measured against the *unconstrained model's* allocation — a baseline with no privileged moral standing, since the constraint exists precisely because that allocation is suspect. On a claims-based view [19], withdrawing an approval that answered to no legitimate claim wrongs no one; a positive classification is also not automatically a benefit (extension in lending can be predatory inclusion). The exchange rate is therefore an unweighted descriptive count, not a welfare ledger: a reader with prioritarian weights on the disadvantaged group's gains may score a rate above 1.0 as net-good, and the group-conditional flows the tables report make such weighted readings computable. What the accounting establishes is narrower and metric-relative: the certifying number cannot tell these outcomes apart, whatever one's weights.

On HMDA mortgage data, however, the direction reverses. The identical constraint *grows* the pool by 4.3%, at an exchange rate of 0.50 — two favourable decisions created for every one destroyed. What makes this worth noticing is that the parity metric cannot see the difference: it reports 0.018 on the population that destroyed a fifth of its favourable decisions, and 0.010 on the one that created more than it destroyed.

V. What Predicts the Direction

A. A single-factor design

ACS Income's label is "earns more than $50,000", and that threshold is a benchmark convention rather than a fact about the world, which makes it a knob we can turn. Moving it on one fixed state varies the selection rate while the sampled people, instrument, feature set and group ratio stay fixed — though moving the label redefines the prediction problem itself, so this is one arm of the two-routes triangulation below, not an isolation of the rate on its own. The change in the pool rises with the baseline selection rate at $r = +0.801$ over four arms — too few points for formal inference, and reported as descriptive; a partial correlation on four points has one degree of freedom and is omitted as uninformative. The sign flip is the substantive observation: 22 decisions are destroyed per one created at a rate of 0.03, compared with 0.75 at 0.89.

B. The transition appears without being manufactured

The same transition also appears where nothing is manipulated at all. Pooling two states across five real loan products, home improvement at a selection rate of 0.555 levels *down* (-1.57%) while refinancing at 0.871 levels *up* ($+2.95\%$), with $r = +0.803$ and Spearman $\rho = +0.900$. This means a bank running one constraint across its loan book would take opportunities away in one product while handing them out in another, and its fairness report would show success in both.

C. It is the rate, not the difficulty of the task

The design above moves the *label*. "Earns over $100,000" is not a rarer version of "earns over $20,000"; it is a harder problem. That design therefore cannot separate two explanations which predict everything observed so far.

We separate them by holding the task completely fixed — same rows, features, groups and *fitted scores* — and moving only where the model draws its line. The relationship reproduces, and so does the crossover's location: on Oregon the two routes give 0.362–0.653 and 0.353–0.637. **Task difficulty is not doing the work.**

With twelve points chosen from each population's viable band, the relationship holds on the Dutch census ($+0.946$), South Carolina ($+0.905$) and COMPAS ($+0.844$). Alabama and Kentucky return *negative* values on eleven and ten arms; their viable bands top out at 0.566 and 0.561, at or below every crossover measured, so both sweeps lie almost entirely on one side of the transition. **We therefore claim the relationship across the crossover and withdraw any claim that it holds within the low-rate region alone.**

D. Five domains, and the group-gap alternative

Fig. 1 shows every arm, and Table IV the summary. The pre-registered alternative — "this is a property of income prediction and American mortgage lending", requiring $|r| < 0.30$ — loses on both new instruments.

TABLE IV
THE RELATIONSHIP ACROSS INSTRUMENTS. EACH r IS COMPUTED WITHIN ONE POPULATION OF THAT INSTRUMENT, OVER THE RETAINED ARMS COUNTED IN THE LAST COLUMN — NOT POOLED ACROSS THE INSTRUMENT, FOR THE REASON FIG. 1 DOES NOT POOL.

Instrument	Domain	Population	r	Arms
ACS Income	income	Alabama	$+0.801$	4
HMDA	mortgages	MS+LA pooled	$+0.858$	4
COMPAS	crim. justice	race arm	$+0.870$	5
Dutch census	occupation	sex arm	$+0.915$	6
LSAC	legal educ.	—	unsweepable	—

TABLE V
WHAT EACH MANIPULATION PRESERVES

Manipulation	Direction	Magnitude
Route to the rate	unchanged	differs $3\times$
Tolerance, $25\times$ range	unchanged	differs $4\times$
Boosted trees (5/5 pops.)	unchanged	differs $3\times$
Attribute, sex $\rightarrow$ race	unchanged	—
Criterion $\rightarrow$ equalized odds	weaker, unevenly	differs $8\times$
Optimiser $\rightarrow$ post-proc.	**vanishes**	—

The oldest competing account is that the *gap between groups* drives the direction. The Dutch census has a gap of 0.298, roughly twice Adult's, making it the population where that explanation should win. It gives $r = +0.915$ across all six arms, none excluded.

Because the parity-gap floor was tuned in-sample (Section III), every row of Table IV was re-scored at floors 0.02/0.05/0.08/0.10: each correlation stays between $+0.80$ and $+0.92$ with no sign change at any floor (HMDA is floor-invariant outright; Alabama thins to two arms only at 0.10). The floor dependence that Table VIII reports for the exploratory states does not reach the domain table.

E. Robustness

Table V summarises. **Within a population, and in the attribute-blind regime, the selection rate predicts which way and orders how much. Neither transfers between populations**: the signed distance from a population's own crossover gives ρ of $+0.96$, $+0.95$ and $+0.78$ in three of four populations, but a pooled slope reaches only $r = +0.487$ against a pre-registered $+0.70$ bar.

VI. A Constraint on One Attribute Leaves Subgroups Behind

A constraint is imposed on the attribute someone chose to name. Kearns *et al.* [6] showed that marginal constraints can be satisfied while structured subgroups are badly treated. Table VI is that failure mode measured on Adult.

Constraining sex drives its gap from 0.190 to 0.020 — a success by the certifying metric. The worst *Sex*$\times$*Race* subgroup gap falls only from 0.315 to 0.178, which is **nine times** the gap that was constrained. The subgroup carrying the residual also *changes*, from Female $\times$ Black to Male $\times$ Black,

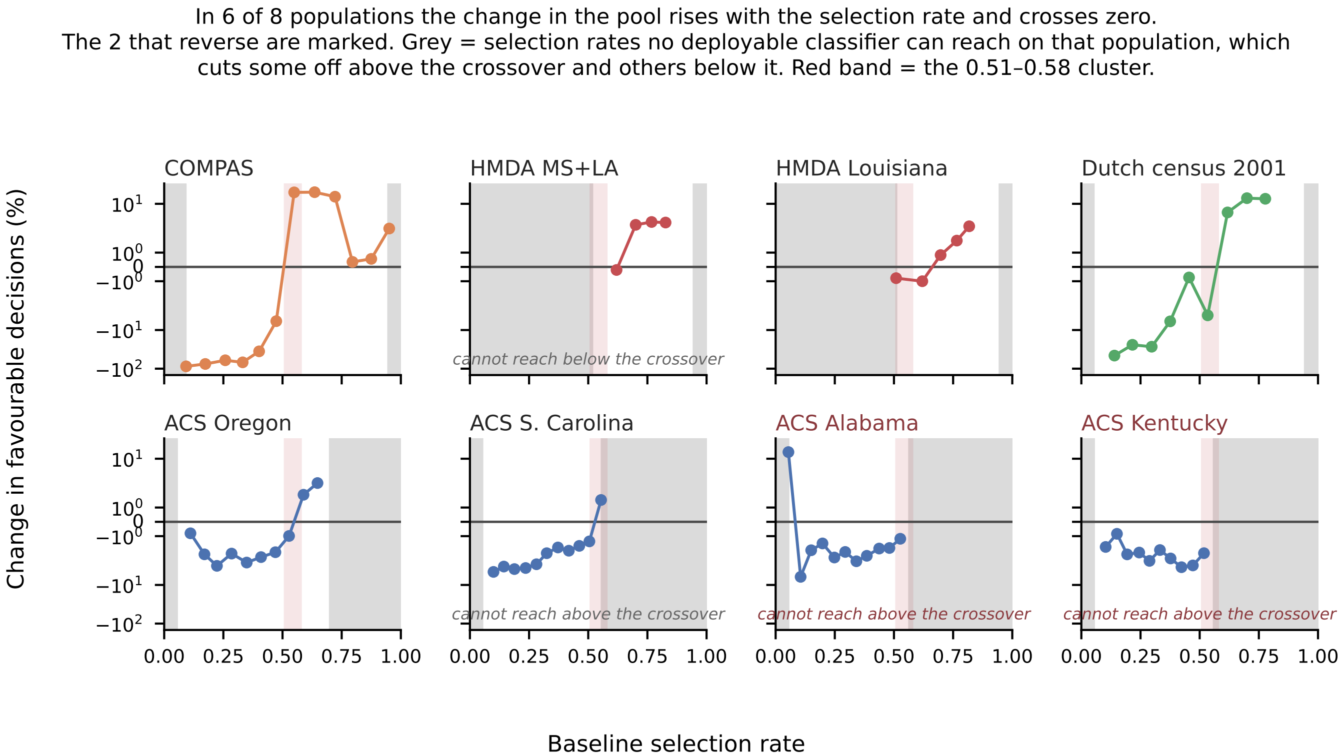

Fig. 1. Every retained arm from every densely swept population. Within a population the change in the pool rises with the baseline selection rate and crosses zero; the two that reverse are marked, and both are sweeps whose viable band confines them below the crossover. Panels are deliberately *not* pooled: across populations a single slope reaches only $r = +0.487$ and fails its pre-registered bar, so one scatter would assert a common curve the data rejects. Vertical band: the 0.511–0.576 cluster of point estimates from Table VII, drawn as first published; its upper anchor (Dutch, 0.576) was later downgraded by the nested bootstrap (Sec. VIII), leaving 0.511–0.558 standing. Carrying the table's bootstrap intervals widens the band to roughly 0.43–0.58.

TABLE VI
UCI ADULT. CONSTRAINING SEX TO 0.020 LEAVES THE WORST *Sex*×*Race* SUBGROUP GAP AT 0.178 — NINE TIMES LARGER — AND MOVES WHICH SUBGROUP IS WORST OFF.

Arm	**Sex**	**Subgroup**	**Worst off**
Unconstrained	0.190	0.315	F×Black
Constrained on sex	**0.020**	**0.178**	M×Black
Constrained on both	0.016	0.048	F×White

so an auditor tracking the originally worst-off cell would see improvement while a different cell absorbed the cost.

This replicates across populations and is more severe than Adult suggests: 13.2× in Mississippi. It is bounded, however, by how large the minority is — the effect scales with minority share at $r = -0.671$, and does not appear at all below a threshold. We present this as an empirical replication of [6] and [4] with a boundary attached, not as a new observation.

A measurement hazard worth naming. Roughly five subgroups per population are too small to support any estimate. A rate with an empty denominator must return undefined rather than zero, or an unmeasurable subgroup masquerades as a perfectly fair one — which is the failure this analysis is most likely to produce and hardest to notice.

VII. THE REGIME BOUNDARY

The last row of Table V looks like a failure of robustness, but we read it differently, because it falls exactly on a line the theory draws. Post-processing reads the protected attribute at prediction time while the reduction does not, and that is the distinction on which [5] is built.

Running post-processing at each population's natural operating point across 17 populations gives $r = -0.024$, compared with $+0.585$ for the reduction on the same populations. Our pre-registered prediction — that the rule would carry over — failed, and its naive alternative won outright, which is what pointed us at the regime distinction in the first place.

In the attribute-aware regime the theory says there is nothing to predict, because the direction is determined. Its prediction for that regime is that the advantaged group loses and the disadvantaged gains, always. On our populations that holds in **18 of 18**, without exception. That check was post-hoc, so we repeated it as a confirmatory test on nine populations never previously post-processed, with the bar set at all nine and committed before the runs: it holds **9 of 9**. One scoring honesty: the coin-flip tail ($0.5^9 \approx 0.002$) is the wrong null to celebrate, because the theorem made the informed prior for this outcome close to one — so this row is a *theory-consistency check* that the pipeline reproduces what the

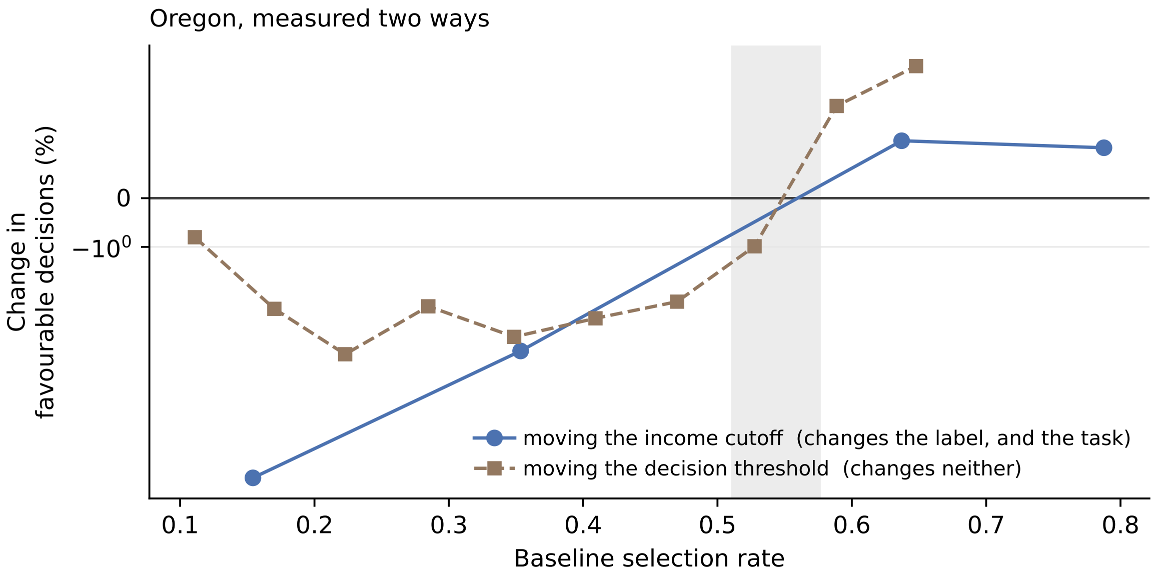

Fig. 2. The design that separates the selection rate from the difficulty of the task. Moving the income cutoff changes the label, and therefore how hard the problem is; moving the decision threshold changes neither, only where the fitted model draws its line. Both routes cross zero inside the same band. If difficulty were doing the work, the second curve would not reproduce the first.

theorem determines, not a forecasting success, and we count it as the former. The regime result therefore stands at 27 of 27, nine of them run to a pre-committed bar. Scoring is strict rather than weak: an arm counts only if the advantaged group's seed-mean losses and the disadvantaged group's seed-mean gains are both positive, and no minimum-magnitude guard was applied — the theorem promises weak inequalities, so a near-zero arm could in principle have scored either way, a rubric gap the later sealed direction tests closed and this count predates.

A word on what we do and do not claim against the theory. Its Theorem 3 states conditions over the extrema of $\zeta(x) = (\eta(x) - c)/\nu(x)$, where $\eta(x)$ is the positive-class probability at x, c the cost threshold, and $\nu(x)$ the group-membership signal — which their paper leaves without a prescribed estimator, so we estimate ν with a logistic probe predicting the protected attribute from the features. **Under that estimator, the extrema orderings could not be certified on any of the 26 arms tested**: the plug-in $\hat{\zeta}$ grows without bound where $\hat{\nu}$ approaches zero, and the two regions' empirical ranges overlapped in every case — on Adult by $[-139, 4.1]$ against $[-39, 6662]$. This is an estimator-relative finding, not a structural one; a different estimator, or the population quantities themselves, might behave differently. The conditions are also *sufficient, not necessary*, so none of this makes the theorem false. It leaves it *unverifiable on these populations as we could measure them*, which is a claim about operational applicability and not a refutation.

The relationship between their quantity and ours turns out to be closer than a horse race. A *relaxed* version of their ordering, replacing the unattainable extrema with 5th/95th percentiles of $\hat{\zeta}$, agrees with the measured direction on **24 of 26** arms, compared with 25 of 26 for the selection rate — and, in a post-hoc comparison, **the baseline selection rate correlates with the relaxed ordering quantity at** $r = +0.935$, the two rules agreeing on 25 of 26 arms. That is, the number this paper uses is not a rival of the theory's condition but a cheap proxy for it: the theory's ordering needs the joint score-and-group distribution and an estimator nobody has prescribed, while the selection rate needs one line of a dashboard, and the two point the same way almost everywhere. The regime distinction itself matters most: the theory says where prediction is possible at all, and the selection rate *operationalizes* its within-regime content — which is something an empirical study can establish and a population-level argument cannot.

VIII. Boundaries

A. The crossover clustered, until more states were measured

Table VII tells the honest history in three blocks. The first four populations, across three domains and two countries, have overlapping intervals, and carrying those intervals gave a cluster spanning roughly 0.43–0.58. Because the table's original intervals resample seeds only — an anti-conservative choice the Setup section concedes — the four located rows were re-run under a nested rows-by-seeds bootstrap (five refitted seeds $\times$ 200 row resamples of the test split, arms rebuilt per seed at fixed target rates). Three survive: COMPAS, South Carolina and Oregon's published locations sit inside their nested 95% intervals (medians 0.457, 0.525, 0.532), with a crossing bracketed in over 97% of resamples. **The Dutch row does not**: under the denser view its mid-band signs interleave and no (seed, resample) pair brackets a crossing at all, so its 0.576 is downgraded to a property of the six arms that produced it, consistent with the non-monotone verdict

TABLE VII

CROSSOVER LOCATION WITH BOOTSTRAP INTERVALS OVER SEEDS (2,000 RESAMPLES). "LOCATED" IS THE FRACTION OF RESAMPLES IN WHICH A SIGN CHANGE COULD BE BRACKETED AT ALL. †DOWNGRADED: UNDER THE NESTED ROWS-BY-SEEDS BOOTSTRAP NO RESAMPLE BRACKETS A CROSSING FOR THE DUTCH CENSUS; ITS ROW IS KEPT AS THE RECORD OF THE ORIGINAL ESTIMATE. THE RATES ARE NOT ONE ECONOMIC OBJECT: SURVEY ROWS ARE PREDICTED-STATUS RATES OVER SAMPLED PERSONS, LENDING ROWS ARE APPROVAL RATES CONDITIONAL ON HAVING APPLIED, AND COMPAS INVERTS A RISK SCORE — COMPARISONS ACROSS BLOCKS CARRY THAT CAVEAT.

Population	Domain	Crossover (95% CI)	Loc.
Non-lending			
COMPAS	crim. justice	0.511 (0.434–0.524)	83%
S. Carolina	income	0.530 (0.526–0.533)	99%
Oregon	income	0.558 (0.556–0.560)	**77%**
Dutch census†	occupation	0.576 (0.572–0.580)	100%
Added by the sealed sweep campaign (bracket, no bootstrap)			
Florida	income	**0.284** (0.259–0.309)	—
Ohio	income	0.556 (0.528–0.585)	—
Pennsylvania	income	**0.652** (0.624–0.681)	—
Lending — one market, three overlapping estimates			
HMDA pooled	mortgages	0.660	—
HMDA Louisiana	mortgages	0.659	—
HMDA refinance	mortgages	0.758	—

the audit itself returns on that population. A sealed sweep campaign across ten further states was then run to test what predicts the location, and the three crossovers it located break the cluster: Ohio lands inside it, while **Florida sits at 0.284 and Pennsylvania at 0.652**, both with tight brackets and every guard passed, on the same instrument the cluster was built from. Located crossovers now span **0.28 to 0.65**, which means the location is population-specific, often but not reliably near the middle. The same campaign found that a third of the large states have no crossover to locate at all: Texas, New York and Illinois sweep U-shaped, with the pool change positive at both window edges and negative between, while New Jersey and Virginia level *up* across their entire windows. The monotone within-population picture, which held on every population swept before, is therefore itself population-dependent.

Every lending estimate sits above every non-lending one, but those come from a *single* market measured three overlapping ways, so "lending is different" remains a hypothesis — and selection into that market's sample is the leading alternative reading: an HMDA approval rate conditions on who applied, which the survey rates do not.

B. The shape boundary, taken cross-task

A twelfth sealed test asked whether the curve-family boundary of the shape seals — LOW below a label base rate of 0.365, HIGH at or above — survives a change of *question*. Six race arms of two folktables tasks this project had never run (employment in three states, public-coverage receipt in three) were sealed with seed-0-fixed thresholds, calls on both sides of the boundary, and the seal's commit externally timestamped before any arm existed. **It scored 0 of 4** (two sweeps returned too few surviving arms to classify), against a bar of scored-1, the constant not beaten. What that settles must be stated carefully, because every scored call was wrong *in the same way*: a perfect inversion is transferred structure with a flipped sign, not the absence of structure, and with two tasks sharing persons this design cannot separate "the boundary is income-specific" from "the boundary's sign depends on the task family" or from a parameterization we have not identified. Two artifacts are ruled out: the family call is computed from pool-change signs, which are invariant to the group declaration, and the labels are folktables' own, so neither of the convention errors our checks guard against can produce this pattern. Recorded beside the failure, without a registered bar: all six populations' natural arms moved *down*, at rates 0.31–0.47 — what the direction rule expects below the crossover, though a constant predicting "down" matches it equally, so it is a consistency observation and no more. The boundary's remaining live test is the IPUMS cohort's S3, on income tasks, real-anchored — the only domain where it has ever called anything correctly.

We therefore treat a crossover near 0.54 as an empirical prior with a measured failure rate, not a constant. It predicted 9 of 10 sealed populations in advance (Section VIII-E); scored post-hoc on the campaign's ten natural arms it manages 5 of 10, and every miss agrees with its own state's sweep — Florida's natural arm turns up at a rate of 0.288, and Florida's located crossover is 0.284. That agreement carries the same caveat as the audit's consistency count: the natural arm is part of the sweep that located the crossover, so miss-agreement is a post-hoc, partially circular check, not independent validation. The within-population relationship survives where the landscape is monotone; the fixed prior transfers to some populations and not others, and nothing in this work yet says which in advance.

The residual question — what predicts the location — was given a sealed test with two candidate predictors, a $\rho \geq 0.70$ bar and a floor of six new locations. It returned UNDERPOWERED: three located, so neither candidate is judged. The earlier screen's two correlations ($+0.724$, $+0.865$, collinear at $+0.947$ over four populations) remain a hypothesis, and any future test must first handle landscapes with no crossover at all.

C. The procedure is unusable on very generous tasks

Varying the selection rate by moving a threshold works by *degrading* the classifier. On a task where most people already qualify, reaching a low selection rate requires a model that refuses qualified people, and past a point that model is worse than a constant. Defining the *viable band* as the range of selection rates reachable by a classifier still beating $\max(p, 1-p)$: Dutch spans 0.895, COMPAS 0.859, typical ACS ≈ 0.51, and LSAC only **0.056**. On LSAC, where 89% pass the bar, no choice of thresholds produces a usable sweep. This is computable before running anything.

When the viable band forbids a sweep, the crossover can still be located by comparing natural sub-populations — which is how the mortgage crossover was first found.

TABLE VIII
SENSITIVITY TO THE PARITY-GAP EXCLUSION. CORRELATION PER POPULATION AS THE THRESHOLD IS LOOSENED.

Population	0.01	0.02	0.05	0.10
Dutch census	$+0.851$	$+0.908$	$+0.946$	$+0.938$
COMPAS	$+0.844$	$+0.844$	$+0.844$	$+0.871$
S. Carolina	$+0.012$	$+0.012$	$+0.905$	$+0.849$
Oregon	$+0.095$	$+0.095$	$+0.664$	—
Alabama	-0.368	-0.368	-0.368	$+0.296$
Kentucky	-0.590	-0.590	-0.654	—

TABLE IX
THE SEALED PREDICTION. POPULATIONS NEVER PREVIOUSLY MEASURED; PREDICTIONS COMMITTED BEFORE THE RUNS.

Population	Rate	Δpool	Pred.	Actual
MO, \$100k	0.026	-16.48%	up	**down**
AZ, \$100k	0.037	-23.34%	up	**down**
IN, \$70k	0.081	-23.32%	up	**down**
TN, \$50k	0.255	-1.68%	down	down
MD, \$50k	0.508	$+0.78\%$	down	**up**
MI, \$30k	0.612	$+0.85\%$	up	up
NC, \$20k	0.789	$+0.16\%$	up	up
GA, \$10k	0.905	$+0.30\%$	up	up
Taiwan 2005	0.885	$+0.68\%$	up	up

D. A sealed prediction, and what it cost

Every result above is a correlation computed after the arms were run, which is the weakest support for a claim of the form *you can know in advance*. We therefore ran the strongest format available: nine populations whose direction had never been measured, with the rule, the populations and the pass mark committed before any of them existed. Eight enter the sealed score; the ninth, Taiwan, was sealed alongside them but registered as reported separately, because a single non-Western population can carry no claim on its own (it was predicted correctly: rate 0.885, observed $+0.68\%$).

It failed. Four of eight, against a bar of seven, not beating a constant.

What failed is a refinement, not the rule. Table VIII shows the exclusion threshold doing real work: South Carolina moves from $+0.905$ to $+0.012$ when it is loosened. Investigating that, we found the newly admitted arms are not noise — at twenty seeds they are consistently *positive*, $+8.73$, $+6.85$, $+9.09$, six standard errors from zero — and concluded the relationship turns back up below a selection rate of 0.10. The sealed rule carried that refinement.

All three held-out low-rate arms came out strongly *negative*. Scoring the same eight arms under the simpler rule we held before the refinement — down below the crossover, up above it — gives **seven of eight**, its only miss being Maryland at 0.508, within 0.032 of the crossover on an effect of $+0.78\%$. That seven is post-hoc and we do not claim it; what was sealed scored four.

The refinement was route-specific. Its arms reach a rate near 0.05 by thresholding one fixed score vector; the sealed arms reach it with a genuinely rarer label. At that rate the two disagree completely, $+8.73$, $+6.85$, $+9.09$ against -16.48, -23.34, -23.32. So Section V-C qualifies: the two routes agree through the middle of the range and diverge at the bottom, and the accuracy rule does not catch the divergent arms — they clear the trivial-predictor bar and remain artifacts.

A refinement drawn from four in-sample populations, internally consistent and backed by a twenty-seed replication, made the prediction worse than the rule it replaced and no better than a constant. Nothing in the in-sample data could have shown that.

TABLE X
THE RE-SEALED PREDICTION: THE UNREFINED RULE — DOWN BELOW 0.54, UP AT OR ABOVE — COMMITTED BEFORE TEN NEVER-MEASURED STATES RAN. BAR: 9 OF 10 *and* BEAT THE BEST CONSTANT.

Population	Rate	Δpool	Pred.	Actual
NE, \$100k	0.014	-6.35%	down	down
AR, \$50k	0.195	-2.55%	down	down
OK, \$50k	0.220	-12.78%	down	down
WA, \$70k	0.233	-5.52%	down	down
NV, \$50k	0.297	-1.57%	down	down
MN, \$30k	0.699	-0.04%	up	**down**
CO, \$30k	0.701	$+0.63\%$	up	up
KS, \$20k	0.767	$+0.58\%$	up	up
WI, \$20k	0.814	$+0.69\%$	up	up
IA, \$10k	0.896	$+0.04\%$	up	up

E. The rule re-sealed, and it holds

After that failure, the only move that could settle anything was to seal the *unrefined* rule — down below 0.54, up at or above, with no low-rate clause — against populations with no measurement history of any kind. We used ten ACS states this project had never touched, one arm per state so that the arm count is the population count by construction, with income cutoffs assigned to spread the expected rates across both sides of the crossover so that no constant could mimic the rule. The 0.54 prior itself predates the test: it is the mid-point of the non-lending cluster, fixed from four populations that include none of the ten. One of those four anchors — the Dutch census — was later downgraded by the nested bootstrap (Section VIII); recomputed from the three surviving locations the mid-point is 0.53, and no call in either sealed cohort, nor in the post-hoc campaign scoring, changes under that shift — the nearest arm to the boundary on any cohort sits at 0.508. The committed prior stands as sealed; its provenance is simply thinner than first stated. The bar was committed with the populations before any arm ran: at least 9 of 10, *and* strictly beating the best constant.

It holds: 9 of 10, best constant 6 (Table X). Two statistics, because they answer different questions and the weaker one is the honest headline. The exact binomial tail — a guesser at the best constant's rate ($p = 0.6$) reaches 9 or more of 10 with probability ≈ 0.046 — answers whether an independent guesser could do this. The *paired* comparison against the best constant on the same ten resolved arms is stricter: rule and constant disagree only on the five arms the rule called *up*, the rule wins four of those five, and the one-sided sign test on the discordant calls gives $p \approx 0.19$. And this is the second sealed attempt at a direction rule after the first failed, so even the binomial tail carries a sequential correction of roughly a factor of two — putting the corrected tail near 0.09, across the conventional line: suggestive, not significant, under the sequential accounting. The pass is evidence, not proof, and the paired test is why we claim no more. A further limit of the cohort's own design must be stated: every arm reaches its rate by the label route, and the achieved rates leave the interior of $(0.30, 0.70)$ unsampled — the nearest arms sit at its very edges (0.297, 0.699, 0.701) — so this pass cannot discriminate the selection rate from label rarity as the operative variable, nor the value 0.54 from any prior in that interval — a rule reading only the income cutoff scores identically here. What the pass establishes is that a mid-band rule predicts direction on unseen ACS-instrument populations; the adequately powered third cohort must therefore be off-instrument, concentrate rates near the crossover, mix both routes, and score against the cutoff-only and 0.5-prior nulls, and it is future work. The prediction used the baseline selection rate and the fixed 0.54 prior alone — no sweep, no per-population characterisation. Below 0.10, where the refuted refinement said up, Nebraska at 0.014 moved down by 6.35%, as the unrefined rule predicts.

The miss is scored by the criterion sealed with it, and it is *not* excused as a boundary call: Minnesota sits at 0.699, far from the crossover, so under the registered rubric it counts against the rule itself. What can be said post-hoc is that its effect is the smallest in the set by an order of magnitude and flips sign across seeds (mean -0.04%) — a rule predicting a sign has nothing to grip on an arm whose true value may not have one, while Colorado, 0.002 away in rate, moved $+0.63\%$ and was called correctly. The registered rubric did not anticipate near-zero magnitudes away from the boundary; a future seal should state a minimum-magnitude guard in advance, and this one is charged the miss for not having done so.

A seed-level accounting sharpens the lesson: 16 of the 19 sealed arms (both cohorts) are unanimous in sign across their five seeds, and the three that split are exactly the near-zero effects. Minnesota, the miss, splits 3 of 5 — and Iowa, at $+0.04\%$, splits 3 of 5 identically *and scored correct*: two statistically indistinguishable arms, one counted for the rule and one against it, which is what scoring near-zero magnitudes means and why Algorithm 1 now refuses them as INDETERMINATE.

One post-hoc robustness check, run because the exclusion rules were tuned in-sample (Section III) and the sealed cohorts are where their choice must not matter: the sealed scoring used no parity-gap exclusion by design, and re-scoring under the audit's own frozen 0.05 floor retains eight of the ten arms at 7 correct against a best constant of 5 — the margin survives. The two arms the floor drops (Iowa and Nebraska, baseline gaps 0.016 and 0.017 at their extreme rates) are arms Algorithm 1 itself would refuse as having too little baseline disparity to audit, which is worth knowing: the cohort's extreme-rate coverage came partly from arms the deployable procedure would screen out.

F. Small populations

Below roughly 2,500 test subjects the method's own randomness exceeds the entire effect of the constraint. This is a specific instance of predictive multiplicity [12], [16] — and Long *et al.* [12] show fairness interventions themselves exacerbate it — models with equivalent aggregate performance disagreeing substantially on individuals — and we position it as a caution supported by that literature rather than as a novel finding. COMPAS has 1,584 and flips sign between seeds on two of four arms; LSAC has 6,240 and flips on none. COMPAS carries direction here and never magnitude.

G. How the withdrawal is implemented: a lottery

The reduction returns a randomized mixture by design [1], and each person's approval probability under that mixture shows which solution the optimiser found. On threshold-sweep arms deep in the score distribution's tail, two qualitatively different solutions appear. Where the constraint *lifts*, the mixture behaves like an informed adjustment: probability is granted below the threshold, rising with the person's score, with the disadvantaged group favoured within each score band. Where it *cuts*, the mixture is a lottery: nobody below the threshold receives any probability at all, and every predicted positive of both groups is kept with one flat probability — 0.65 on the Dutch census arm, 0.135 on COMPAS, where a person's keep probability is constant in their own score (the correlation is degenerate: score carries no information about who is kept). The gap closes because discarding a fixed fraction of everyone's approvals at random shrinks it multiplicatively; the certificate reports the same success either way. A single quantity — the mean probability granted just below the threshold — separates the two behaviours on all nine deep-tail arms tested, including a matched pair that agrees on every aggregate we can compute and moves in opposite directions.

Because the nine arms above sit in the region our own limitations call unreliable, the same probe was run as a control at the *natural* operating point of seven populations spanning both directions — Adult, three cutting and three extending survey populations. **The lottery does not appear at any of them**: every natural-point mixture is graded, granting probability below the boundary (means 0.17–0.43 against 0.00–0.07 on the deep cut arms) with keep probability rising in the person's own score (correlations $+0.37$ to $+0.73$ against ≈ 0), even on Adult, whose constraint removes a fifth of the pool. The signature is therefore a property of constraints pushed to severe operating points, not of withdrawal as such — which narrows the hazard and sharpens the audit: a flat keep probability in one's own fitted mixture is not the expected cost of parity but a flag that the optimiser has stopped using the scores, and the quantity that detects it is computable from any fitted mitigation.

Three delineations keep this claim its right size. In the *attribute-aware* setting, the optimal randomized fair classifier is characterized as randomizing at group-specific decision boundaries [13]; the attribute-blind analogue is not characterized, and a flat keep-probability is what a two-member blind mixture produces when one member grants nothing — which is the observed structure here; two distinct-threshold members would instead produce a rising step — so the lottery differs in form from the aware-regime optimum. Whether it is suboptimal *within* the blind constrained class was an open computation; we ran it. Searching the minimal richer class — two-threshold blind mixtures, with the parity weight in closed form and the threshold grid extended into the 0.9975 score quantile to rule out a grid-edge artifact, with the pool matched to ± 0.005 and the parity gap capped at the lottery's own — **the feasible set is empty on all ten seed-arm pairs**: no score-informed blind mixture *of that two-threshold form* attains the lottery's parity gap and pool size at either signature arm. Within the class searched, then, the lottery is not an optimiser failure; richer blind classes (more thresholds, mixtures over refits) remain unsearched, so the claim is family-relative and stated as such. What it licenses is narrower and still useful: the obvious repair a reviewer would propose does not exist, which is exactly why the honest charge is the missing announcement, not the randomization. The known arbitrariness of fair models is *across* retrained models [12]; this is deliberate randomization inside one fitted model. And randomness itself has been advocated as a fairness device [15], and the philosophical literature holds that a lottery among relevantly equal claims can be exactly what fairness requires [19], [20] — so the charge here is deliberately narrower than "a lottery occurred": it is that a lottery occurred *unannounced*, under a certificate indistinguishable from an informed adjustment, in settings where individualized reasons are owed. Accepted allocation lotteries are public about being lotteries; this one is certified as fairness. The standard derandomization remedy [14] applied to a flat lottery amounts to re-choosing a threshold, which undoes the parity the lottery bought; for an auditor, the flat-probability signature itself is the usable part, and it is computable from any fitted mitigation.

IX. What to Do About It

Algorithm 1 states the procedure. Three of its steps are worth drawing out, because each is a place where a plausible-looking run returns a meaningless answer.

It can refuse. Lines 6–8 compute the viable band and exit before any constraint is fitted, if no threshold reaches a low selection rate with a model still beating the trivial predictor. On LSAC that band spans 0.056 and the procedure declines, rather than returning arms that would be discarded later.

It can return VOID. Line 16 requires enough arms to survive the guards and the pool to have moved at all. A correlation computed over arms that barely move is fitted to noise: Connecticut returns $r = -0.924$ on a spread of a tenth of a percentage point, and without this test that number reads as a refutation of the whole finding.

It can return NON-MONOTONE, and on current data it often will. Lines 20–22 exist because the assumption behind bracketing — one rising sign change — is itself population-dependent: among the ten largest states we swept, three are U-shaped and the 2022 vintages at nominal labels invert, so a bracket step that assumed monotonicity would return a confident wrong direction precisely there. A NON-MONOTONE verdict means the directional rule does not apply to this system, which is a first-class answer, not a failure of the audit. A sweep whose retained arms all share one sign is the opposite case and returns that direction (line 24): the constraint moves the pool the same way across the whole viable band, which is what "levels up throughout" means operationally. Two cautions the earlier version of this algorithm kept in prose are now steps: arms below a selection rate of 0.10 are marked *advisory* and excluded from the shape call (line 14), because below that boundary the two routes to a rate give opposite signs (Section VIII) and the accuracy guard does not catch it; and every fitted arm is probed for the flat keep-probability signature (line 19), because a lottery under the certificate is a finding the direction verdict alone would hide. The probe's decision rule, stated so it can be implemented: flag an arm whose mean granted probability just below the boundary is under 0.10 while the keep-probability's correlation with the score above it is ≈ 0 — the observed separation is wide (0.00–0.07 against 0.17–0.43, and ≈ 0 against $+0.37$ to $+0.73$). Provenance of the remaining guards, since tuned constants are where exclusions hide: the 0.40 band span, the 2-point void guard, the 12-threshold grid and the 0.10 advisory boundary all come from the exploratory phase, like the two exclusion rules whose provenance Section III discloses; all are frozen verbatim in the third cohort's committed protocol, and "within seed noise" in line 28 is the 1.0-point magnitude guard that protocol states.

It can return INDETERMINATE. Lines 28–29 refuse a sign call when the deployed rate sits inside the located bracket, or when the nearest arms' effects are within seed noise — precisely the two conditions under which this project's own record shows a sign call has nothing to grip: Maryland missed at 0.508 at the cluster's edge, and Minnesota's miss was an effect of -0.04% that flips sign across seeds. The registered rubric of the re-seal did not have this verdict and was charged a miss for it; the algorithm inherits the lesson.

It returns a sign, never a size. The distance from the crossover *orders* the effect within a population, so a team can rank its own segments; but no slope transfers between populations, so lines 32 and 34 return a direction and stop there.

What it needs, and what it costs. The audit requires a held-out sample carrying the protected attribute: audit-time access to a, which is legally and practically distinct from decision-time use and is the regime bias-testing provisions contemplate (Section I-A) — in US credit, the self-test privilege of Regulation B (12 CFR §1002.15) exists for exactly this, though it is conditional: the privilege attaches to a lender's own voluntary self-test and expects appropriate corrective action, and an analysis of public HMDA records, like this paper's, is not itself a privileged self-test. It fits the constraint $12 \times \text{seeds}$

times; with five seeds and the linear learners used here, one fit takes 2–10 seconds at 15k–69k training rows (measured on a laptop CPU), so a full audit is minutes per population, and its cost scales linearly in the base learner's own fit time since each reduction call is a sequence of learner refits. A cheap pre-screen exists: the relaxed-ζ ordering of Section VII needs no constraint fits and correlates with the predictor at $r = +0.935$.

What it actually returns, totalled. Run over the 45 population-label pairs with stored sweeps in this project (populations never swept cannot enter), the distribution is: 27 directional verdicts (20 WITHDRAWAL, 7 EXTENSION), 6 NON-MONOTONE, 10 VOID, one refusal at the noise floor (COMPAS), and one bracket with no natural arm to read against. Every one of the 27 directional verdicts matches what the fitted constraint did at that population's natural operating point — a consistency check on the audit's reading of its own sweep, not independent validation, since the natural arm is part of the curve being read. The honest summary: on this project's own data the audit answers six times in ten, and refuses or reports non-monotonicity the rest — a procedure that mostly answers, with its refusals totalled rather than hidden. Finally, an ex-ante verdict earns trust only if checked: whatever the verdict, including refusals, the deployed system's measured pool change and who-pays decomposition (Section IV-A) should be reported after the fact — the verdict is falsifiable at deployment, and Section XII makes this a requirement rather than advice.

The direction survives the extraction a deployer must apply. Every direction result above is measured on the randomized mixture, but a lender under adverse-action-notice duties deploys a *deterministic* model, so we tested the extraction directly: thresholding the mixture's per-person probability at 0.5 on the natural arms of ten populations spanning both directions. The extracted model agrees with the mixture's pool direction on **10 of 10**, with the parity repair largely intact (extracted gaps 0.009–0.034 against baseline gaps 0.08–0.19). One deployment fact surfaced on the way: an extraction that instead tries to reproduce the mixture's own operating point cannot generally do so, because the mixture's probability takes few discrete values and the ties force the deterministic threshold off the target rate — on three cutting populations far enough off to flip the sign of the measured pool change. Thresholding at 0.5 is the well-defined extraction, and it is the one that preserves the verdict. The caveat composes with Section VIII-G: these are natural arms, where the mixture is graded; a flat-lottery arm has no score-respecting extraction, and derandomizing it undoes the parity it bought.

The floor is **not a new method**: a selection-rate floor is a linear moment constraint inside the framework of [1], and a variant of the minimum rate constraints of [2]. It costs about 0.12 accuracy points and moves the exchange rate from 1.47 favourable decisions destroyed per one created to 0.88 across nineteen arms, landing below 1.0 in sixteen of them, with the benefit scaling with the damage at $r \approx -0.99$ — sub-1.0 being a descriptive count of predicted labels, not a welfare verdict, per the accounting caveat of Section IV-A; the group-conditional flows are what a welfare reading would weigh. And because magnitude does not transfer across populations, the floor is issued provisionally: the ex-post measurement the Discussion requires — the deployed system's pool change and who-pays decomposition, with and without the floor — is part of the prescription, not an optional follow-up.

X. Method, and What We Got Wrong

Every prediction was registered with its thresholds *and the naive alternative it had to beat*, before the data existed. Table XI is the complete ledger — every registered or sealed test this work ran, with its committed bar and its outcome, failures included and uncorrected. The rows should not be read as twelve attempts at one claim, of which two succeeded: each test targeted a *different* claim, each failure removed a stronger claim than the one that survives, and the last column records what each row settled. The failures are boundary measurements; the passes are tests of the claim *as those boundaries define it*. Three episodes matter for reading the results above.

A held-out prediction, scored as a failure. The clearest evidence format available is a prediction registered before the data exists. We ran one: after adding the accuracy rule below, four HMDA populations the sweep had never touched were swept and scored under it, with the prediction that each would show the relationship, clear the spread guard and bracket a crossover. **Two of four did**, so the test is scored a failure. Its named alternative — that lending is a domain where this route does not work — nonetheless loses on all four, with $r \geq +0.80$ throughout and $+0.995$ on refinance. Three overlapping estimates of the lending crossover agree (Table VII).

A pre-registered test that failed. The rule does not survive equalized odds at the bar we set: $r = +0.644$ on two states, $+0.334$ on five, against a $+0.70$ threshold. Alabama alone would have given $+0.822$ with every prediction holding.

Two headline results withdrawn. The operating-point design varies the selection rate by degrading the classifier, and nothing bounded how far. Excluding arms beaten by the trivial predictor leaves Alabama with two arms and LSAC with one: $r = +0.979$ and $r = +0.968$, both previously reported, are void. The same exclusion *rescued* mortgage lending, which had failed outright.

A derivation that a constant beat. The most obvious extension of this work is a theoretical account of *why* the selection rate should matter. A general theory already exists and we cite it: [5]'s Theorem 3 gives explicit conditions, over the extrema of the score-region quantity $\zeta(x) = (\eta(x) - c)/\nu(x)$, under which every group's rate weakly falls or weakly rises. Those conditions are stated over population score-region extrema that an empiricist must estimate, and Section VII reports that under the one estimator we tried they could not be certified on any of 26 arms — while a relaxed form of the same ordering tracks the direction on 24 of 26, statistically indistinguishable from the selection rate's 25 of 26.

We attempted the missing step — a derivation predicting the sign from the selection rate directly — and pre-registered

TABLE XI
EVERY REGISTERED OR SEALED TEST. EACH ROW TARGETS A DIFFERENT CLAIM; THE LAST COLUMN IS WHAT ITS OUTCOME SETTLED. NOTHING IS OMITTED.

Test	**Bar / outcome**	**What it settled**
HMDA sweep	4 of 4; got 2 — fail	our sweep procedure is unreliable on mortgage data, so no claim rests on it there
EO transfer	$r \geq +0.70$; $+0.334$ — fail	the rule works for parity, which controls approval rates directly; it failed for equalized odds, which does not
Post-processing transfer	carry over; $r = -0.024$ — fail	the rule stops where the model may read the protected attribute — the boundary of Sec. VII
Derivation	beat a constant; beaten — fail	we cannot explain *why* the rule works, and claim only that it does
Sealed rule, refined	7 of 8; got 4 — fail	the extra clause added from early data was wrong, and is withdrawn
Attribute-aware consistency	9 of 9; got 9 — **holds**	where the attribute is readable, the disadvantaged group always gains; a theory-consistency check run to a pre-committed bar, not forecasting skill — the theorem made the prior ≈ 1
Re-seal, unrefined rule	9 of 10 + beat constant; got 9, constant 6 — **holds**	the simple rule predicts populations it has never seen, from the rate alone
Crossover residual	6+ located, $\rho \geq 0.70$; 3 located — underpowered	neither candidate judged; a third of large states have no crossover to locate
Sealed magnitude model	beat predict-zero; MAE 4.50 v. 0.77 — fail	size cannot be predicted from distance and band span; the direction-only concession is earned
Sealed shape boundary	scored-1 + beat const.; 4 of 6 — fail	the base-rate boundary called every 2014 shape and both sealed flips; the 2022 inversion was since traced to the nominal label sliding (see Limitations)
Sealed attribute-independence	scored-1 + beat const.; 2 of 6 — fail	shape is not a label-side property: every race arm is classic at tenfold the sex arms' signal, pointing at gap size as the governing variable
Sealed cross-task shape	scored-1 + beat const.; 0 of 4 — fail	the boundary does not transfer to two non-income tasks; every scored call inverted — transferred structure with a flipped sign, unexplained (Sec. VIII-B)

its bars. **It cleared them and was then beaten by a constant**, 12 correct against 13 out of 14 held-out arms. That failure is why this paper reports an empirical regularity rather than a mechanism, and why we do not attempt a fourth derivation here.

A plausible result that was arithmetic. A post-processing comparison appeared to show a clean reversal. It was void: the method re-derives its own thresholds and returned an identical model at all six operating points, so the measured effect was a constant divided by a shrinking baseline. What exposed it was an exchange rate of 0.000.

The claim that survives every test above, stated once and in full. Within a population whose response is monotone, in the attribute-blind in-processing regime, across the transition: the baseline selection rate, read against *that population's* crossover, predicts the direction in which a parity constraint moves the pool of favourable decisions, and orders the magnitude within that population. The fixed 0.54 prior is a fallible substitute for the population's own crossover: sealed at 9 of 10 in advance on one cohort of unseen populations, 5 of 10 post-hoc on a second cohort of larger, richer states whose misses each agree with their own measured crossovers. The boundaries, equally part of the claim: located crossovers span 0.28–0.65 and nothing yet predicts the location; some large populations have no crossover at all, sweeping U-shaped or levelling up throughout; magnitude does not transfer across populations and a sealed model of it failed; the deepest-tail arms of a threshold sweep are unreliable in direction; and in the attribute-aware regime the direction is determined by theorem, the rate predicts nothing, and there is nothing left to predict.

XI. LIMITATIONS

The sealed record is five failures, one pass, and one underpowered test (plus the attribute-aware consistency check). The first sealed test (Section VIII-D) failed at four of eight, carrying an in-sample refinement; the re-seal of the unrefined rule (Section VIII-E) passed at nine of ten, beating the constant at six; a sealed magnitude model then failed outright, and the sealed residual test returned no verdict. The pass's own limit turned out to bind sooner than expected: its ten populations were mid-size, mid-income states, and when ten larger, richer states were swept for the residual campaign, their natural arms scored the fixed prior at five of ten post-hoc — with every miss agreeing with its own state's measured crossover, so the within-population claim held while the transferable prior narrowed. The single miss in the sealed pass is charged to the rule by the registered rubric, because no minimum-magnitude guard was stated in advance.

The two manipulations diverge in the score tail. The deepest-tail arms of an operating-point sweep are unreliable in direction at whatever rate they land (observed from 0.027 to 0.142): they close the gap by *lifting* where every natural-route arm at a matched rate closes it by *cutting*, and the accuracy rule does not separate them — all fifteen such arms clear it. What each solution looks like inside the fitted model is Section VIII-G; what *selects* lift over cut is measured to be none of the rate, the gap, the qualified reservoir, or the threshold height, and remains open. We claim the relationship across the transition, on the label route and natural subpopulations.

Most of the evidence is about predicted labels, not allocations. In five of the seven data sources no one receives anything: "favourable decisions" are a classifier's labels over people whose incomes, occupations or bar results already exist, so the pool being given and taken is a pool of predictions. HMDA is the one source recording real allocative decisions — and it is also where the direction reverses and where our sweep protocol failed its held-out test, so whether the construct transports from status-prediction benchmarks to allocation data is an open question, not an assumption this paper is entitled to. The practical advice of Section IX is addressed to teams making real allocations; they should read this limitation first. The 2022 episode below doubles as the same warning from the label side: a crossover is a property of a (population, real-anchored label) pair, not of a state's name.

Magnitude does not transfer between populations, only within them — and a sealed model of it lost to predicting zero. **The crossover's location is unexplained and wider than it looked**: located values span 0.28–0.65, the sealed test of the two candidate predictors returned underpowered at three locations, and some populations have no crossover to locate. **Coverage** is seven data sources, three countries and two decades; Taiwan is the only non-Western population, and one population is not coverage. The only EU population is the 2001 Dutch census, which by this paper's own vintage standard grounds no claim about current EU deployments. **The regime result** was post-hoc at 18 of 18 and has since been confirmed pre-registered at 9 of 9, but it confirms a published theorem rather than establishing one. **The remedy is a variant**, not a discovery. **And every claim carries a data vintage**: all populations predate 2020 except four 2022 state-years, and those four *invert* the within-population relationship — positive at low rates, negative at high, a shape absent from every earlier vintage. Diagnosis then ran three candidate mechanisms to ground. The survey's recoded relationship column is acquitted by two 2019 sweeps that carry it and behave classically; pandemic-era nonresponse weighting is acquitted at label level, since applying the Bureau's `PERWT` corrections moves 2022's rates and gaps no more than 2018's; and the third candidate is **confirmed as the substantial cause**: re-sweeping the four states with the label at $60,000 — the real-equivalent of 2018's $50,000, landing their base rates exactly on the 2018 values — restores Ohio to a clean classic crossing at its old location and removes Alabama's and South Carolina's inverted limbs entirely, leaving only Nevada as a below-guard residual. The "inversion" was therefore substantially the fixed nominal label sliding down an inflated, bottom-compressed earnings distribution: the task changed under the label, and measured at constant real value the relationship shows no detected time-dependence from 2014 through 2022. The lesson replaces the caveat: fixed nominal outcome definitions are a different task each vintage, so labels should be real-anchored and deployments should audit their own current data.

XII. Discussion

The finding a practitioner can use is small and specific: **run the audit of Section IX on your own current data — it will locate your crossover, or tell you that your system has none — and where it returns a direction, the rate predicts whether a parity constraint is about to give or to take.** The sweep itself is the vintage test: the 2022 episode — an apparent inversion traced to a nominal label sliding down an inflated income distribution — shows that no prior fitted elsewhere, including 0.54, substitutes for measuring one's own current data, and an inverted or non-monotone sweep is a finding about your system, often about your *label*, not a failed audit. Re-run it whenever the model, the population, or the real value of the outcome definition changes. We do not claim that the selection rate *causes* the direction; we claim that it predicts it, within a population, in the attribute-blind regime, and with the boundaries of Section VIII attached.

What makes this worth stating is not that the rule is surprising once seen, because it is not. It is that the certifying metric — the number a deployment team reports, an auditor checks, and regulators in some regimes already require — comes out identical in both directions. A constraint that withdraws a fifth of all favourable decisions and one that creates more than it destroys produce the same certificate, so nothing in the existing reporting pipeline would ever distinguish them.

The literature already knew the metric was silent about this. What was missing, and what this paper adds, is a way to know in advance — from a number a team already has — which of the two outcomes its own system is heading toward.

A procedure meant for practice must also be stated adversarially, because its inputs belong to the party being audited. The label definition sets the selection rate (our own single-factor design moves it across $[0.03, 0.9]$); the population segmentation decides which sub-populations are audited at all, and the HMDA purposes show one constraint earning "extension" on the pooled book while concentrating withdrawal in one product; the sweep parameters and exclusion floors were shown above to move retained correlations; and the refusal verdicts could be engineered as exits. The countermeasures are the ones this project already practices on itself: the audit unit is the deployment decision unit, fixed and disclosed before results are computed; the sweep protocol — grid, seeds, tolerance, baseline model, exclusion floors, real-anchored outcome definition — is committed before the sweep, as our seals were; results are reported at the pooled level *and* for every sub-population the deployer prices or markets separately; and every verdict, including a refusal, is paired with the ex-post measured pool change and who-pays decomposition of the deployed system, so that escaping the ex-ante audit buys no escape from the accounting. Without those commitments the direction verdict would be one more gameable certificate, which is the failure mode this paper exists to name.

Three implications reach beyond deployment teams, each following from measurement rather than doctrine. For certification and audit regimes: because a gap statistic certifies withdrawal and extension identically, any mandated parity or disparate-impact threshold would be materially more informative if it required reporting the pool change and the who-pays decomposition alongside the gap — a reporting recommendation only, requiring no doctrinal commitment, and one with an immediate home: for credit and employment systems the EU AI Act already requires documented bias examination, accuracy and robustness documentation, and post-market monitoring, and the audit's verdicts — including its refusals — are exactly the kind of artefact those files are meant to hold. For notice obligations: a flat-lottery model (Section VIII-G) cannot generate truthful individualized reasons for a denial, since the individual's own score played no role in it, which sits uneasily with adverse-action-notice duties wherever they apply — in the EU, a decision randomized independently of the person's own score is also hard to square with the "meaningful information about the logic involved" that GDPR Articles 13–15 and 22 demand of automated credit decisions — and unlike accepted lotteries in visa or school allocation, this one is not announced as a lottery but certified as fairness. The natural-point control of Section VIII-G bounds the worry: the lottery appeared only at severe operating points, so it is a checkable hazard, not the default cost of parity. One implication is addressed to toolkit maintainers directly: the reduction's fitted mixture already exposes everything the probe needs (its member predictors and weights), so a warning on degenerate support — a member that grants nothing, a keep probability flat in the score — is implementable in a few lines, and would surface at fit time the situation this paper had to discover by audit. And a tension worth naming for law-and-policy readers without advocating anything unlawful: the attribute-aware regime the law disfavours is precisely the one where our data show the intervention's effect is determined and always favourable to the disadvantaged group (27 of 27), while the legally favoured blind regime is where levelling down and the lottery live. The deontic, process-based rationale for blindness is untouched by any outcome measurement; what these measurements show is narrower and still worth saying: blindness is not outcome-neutral.

Appendix: Two Worked Traces of Algorithm 1

Oregon 2018, sex, to a verdict. Demographic parity constrains selection rates: in scope (line 1). Trivial-predictor accuracy $\pi = 0.636$ (line 4); the viable band clears the 0.40 span bar, so the sweep proceeds (lines 6–9). The stored sweep holds 18 arms; the guards discard 2 on the parity-gap floor and 2 as worse than doing nothing, leaving 14 spanning rates 0.111–0.653, none below the 0.10 advisory boundary (lines 11–15). Their pool changes span 7.14 points, clearing the void guard (line 16). The signs, sorted by rate, are −−−−−−−−−−−+++: one rising change, so the landscape is monotone-crossing (lines 20–25) and the bracket is $c = (0.528, 0.589)$ (line 26). The deployed model's rate is $r_0 = 0.353$ — below $\min c$, outside seed noise: verdict WITHDRAWAL, add a floor (line 32). What the constraint in fact does at Oregon's natural arm: -3.0% of the pool.

LSAC bar passage, to a refusal. Trivial-predictor accuracy $\pi = 0.890$ — almost everyone passes. The viable band spans 0.056, far under 0.40: verdict UNSWEEPABLE at line 8, before a single constraint is fitted. Had the guard been ignored, the stored sweep shows what awaited: five of seven arms fall below the trivial predictor and the two survivors are too few to read — VOID. Both endings refuse; neither manufactures a direction.

Reproducibility

Every load-bearing figure is re-derived from stored results by an automated check that fails if a document and its data disagree; 55 checks across five suites accompany this work, 28 of them of that document-against-data form, runnable as `python -m tests.`*`suite`* (or `python -m pytest tests/`) from the repository: `github.com/Dheirav/who-pays-for-fairness`. Pre-registrations are committed before the runs they govern, and every row of Table XI has a registration and a scoring commit, listed here as (sealed → scored): HMDA sweep `e23841b` → `4f9f715`; EO transfer `e658ed1` → `3baa054`; post-processing transfer `bf36db0` → `c097dba`; derivation `a859d16` → `2b66a97`; refined rule `f64d9a7` → `16e2470`; attribute-aware `6660b06` → `7b3609a`; re-seal `356bfa5` → `f1eed99`; crossover residual `93c3446` and magnitude `f0f8056`, both scored in `7b6764f`; cross-year shape `391cd17` → `e6cdb68`; attribute independence `aa6532b` → `d3bcb54`; cross-task shape `d8bfae8` → `aa8d43a`, the first seal whose hash is also anchored externally (an OpenTimestamps receipt committed in `research/seals/` before any of its arms ran) — each seal a committed analyser whose population lists an isolation guard forbids changing. An honest limit of this apparatus: the ordering rests on git commit timestamps from the authors' machine, which an outsider must take on trust. Two seals no longer do: the cross-task seal (`d8bfae8`) and the third cohort's stage-A protocol (`50d467f`) each carry an OpenTimestamps receipt in `research/seals/`, committed before their arms ran and since upgraded to Bitcoin block attestations, so their chronology depends on a public blockchain rather than our clock. The ten earlier seals predate this practice and rest permanently on commit timestamps; that cannot be repaired retroactively, and every seal from stage A onward is anchored at sealing time. A checksum manifest of every input data file (`research/data-manifest.csv`) pins the bytes each result was computed from, because ACS PUMS files are revised in place and the 2022 episode is this paper's own evidence that a different vintage is a different task. Experiments ran on Python 3.12 with scikit-learn and Fairlearn per the repository's pinned requirements; all seven data sources are public.

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Algorithm 1 Directional audit: decide whether the predictor applies at all; if it does, locate the crossover and decide whether to add a floor

Require: deployed model h, held-out (X, y, a) *of the vintage to be deployed on*, fairness constraint C

Ensure: predicted direction, and whether a selection-rate floor is needed

if C does not constrain selection rates **then**
  return OUT OF SCOPE {e.g. equalized odds}
end if
$\pi \leftarrow \max(\bar{y},\, 1-\bar{y})$ {trivial-predictor accuracy}
$s \leftarrow h(X)$
$\mathcal{V} \leftarrow \{\tau : \mathrm{acc}(s>\tau) \geq \pi \ \wedge\ 0.05 \leq \overline{(s>\tau)} \leq 0.95\}$
if $\mathrm{span}(\mathcal{V}) < 0.40$ **then**
  return UNSWEEPABLE; compare natural sub-populations
end if
$T \leftarrow 12$ thresholds from $\mathcal{V}$, spread evenly in *selection rate*
for all $\tau \in T$ **do**
  $r_\tau \leftarrow \overline{(s>\tau)}$; fit C on the arm thresholded at τ
  $\Delta_\tau \leftarrow \%$ change in favourable decisions
  discard τ if parity gap < 0.05 or acc $< \pi$; mark τ *advisory* if $r_\tau < 0.10$ {route-divergence region}
end for
if fewer than 4 non-advisory arms remain, **or** $\max\Delta - \min\Delta < 2$ points **then**
  return VOID {nothing moved; not a refutation}
end if
probe each fitted arm’s mixture for a flat keep-probability (Sec. VIII-G); flag any flat arm
if $\mathrm{sign}(\Delta_\tau)$ changes more than once across the non-advisory arms, **or** changes from $+$ to $-$ as r_τ rises **then**
  return NON-MONOTONE {the rule does not apply here}
end if
if every non-advisory Δ_τ shares one sign **then**
  return that sign’s verdict {it holds across the whole viable band}
end if
$c \leftarrow \big(\max\{r_\tau : \Delta_\tau < 0\},\ \min\{r_\tau : \Delta_\tau > 0\}\big)$
$r_0 \leftarrow$ selection rate of the deployed model
if $\min c < r_0 < \max c$, **or** $|\Delta|$ at the arms nearest r_0 is within seed noise **then**
  return INDETERMINATE {the sealed record’s misses live here}
end if
if $r_0 \leq \min c$ **then**
  return WITHDRAWAL; add a selection-rate floor
else
  return EXTENSION; the floor buys little
end if